Sumukha R Bharadwaj

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Research Interests

Circumgalactic Medium (CGM), Intergalactic Medium (IGM), Quasar Absorption Line Spectroscopy, Galaxy–Absorber Connections

Education

2024–Present: **Ph.D. in Astronomy and Astrophysics**
Indian Institute of Space Science and Technology (IIST), Thiruvananthapuram
Advisor: Dr Anand Narayanan

2021–2023: **M.Sc. Physics**
St. Joseph's College (Autonomous), Bengaluru City University
SGPA: 8.25

2018–2021: **B.Sc. Physics, Chemistry, Mathematics**
St. Joseph's College (Autonomous), Bengaluru City University
SGPA: 8.2

Examinations Qualified

GATE 2024: All India Rank (AIR): 1605
GATE Score: 430

Computational Skills

Programming: Python (scientific computing, astrophysical data analysis)

Astronomical Software: IRAF, PyRAF

Mathematics: Maxima

Typesetting: LaTeX

Research Experience

PhD Research: Spectroscopic study of the Circumgalactic Medium (CGM) and Intergalactic Medium (IGM) using quasar absorption line systems, combining HST–COS spectroscopy with VLT–MUSE observations to investigate galaxy–absorber connections and gas kinematics around galaxies.

Master's Dissertation: *Transiting Exoplanetary Atmospheres – Spectroscopic Reduction and Observational Techniques*
Indian Institute of Astrophysics (IIA), Bengaluru
Supervisor: Dr Sivarani Thirupathi

Publications

S. R. Bharadwaj et al.,
A Partial Lyman Limit Absorber Tracing Cold-Mode Accretion,
Submitted to The Astrophysical Journal (under review).

Workshops and Summer Schools

2023: **Aditya-L1 Workshop**, ISRO in collaboration with St. Joseph's University, Bengaluru. Participated in workshop sessions and served as part of the organizing team.

Dec 2024: **Manipal Astrostatistics School**, Manipal University (Dec 10–15, 2024)

Jan 2025: **Scientific Grant Proposal Writing Workshop**, IISER Thiruvananthapuram

Oct 2025: **COSPAR JWST–UVIT Hands-on Workshop**, Centre of Excellence in Astronomy and Astrophysics, Department of Physics and Electronics, CHRIST (Deemed to be University), Bengaluru (Oct 20–31, 2025)

Professional Experience

Project Assistant: COSMOS Outreach Programme, Indian Institute of Astrophysics (IIA). Participated in astronomy outreach activities including public lectures, observational demonstrations, and educational programs.

Teaching: Home tutoring for high school and pre-university students in physics and mathematics.

Teaching and Outreach

Conducted astronomy outreach and night sky observation sessions for students and the public using outreach telescopes.

Awards and Achievements

State Award for Best Child Actor for performance in a drama production.

Other Interests

Hindustani Classical Music (Praveshika Poorna)

Acting in Kannada films, television serials, and documentary productions.